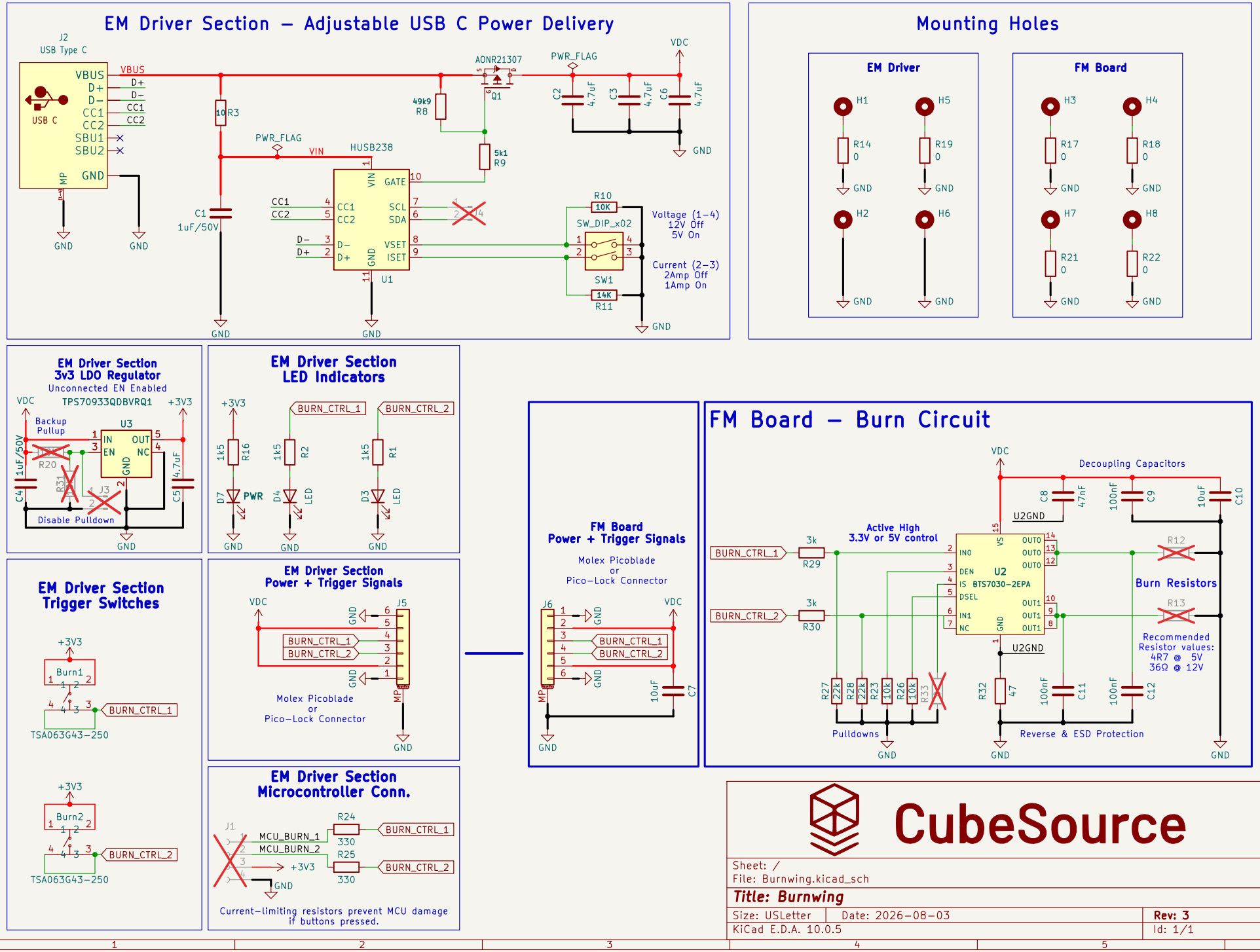

EM Driver Section – Adjustable USB C Power Delivery
J2
USB Type C
USB C
VBUS
D+
D–
CC1
CC2
SBU1
SBU2
GND
MP
VBUS
10 R3
C1
1uF/50V
PWR_FLAG
VIN
HUSB238
AONR21307
Q1
49k9
R8
5k1
R9
PWR_FLAG
VDC
C2
4.7uF
C3
4.7uF
C6
4.7uF
GND
CC1
CC2
D–
D+
VIN
GATE
SCL
SDA
VSET
ISET
GND
U1
R10
10K
SW_DIP_x02
SW1
14K
R11
Voltage (1–4)
12V Off
5V On
Current (2–3)
2Amp Off
1Amp On
Mounting Holes
EM Driver
H1
H5
H2
H6
R14
R19
0
FM Board
H3
H4
H7
H8
R17
R18
R21
R22
EM Driver Section
3v3 LDO Regulator
Unconnected EN Enabled
TPS70933QDBVRQ1
U3
Backup
Pullup
R20
R31
J3
Disable Pulldown
C4
1uF/50V
C5
4.7uF
+3V3
EM Driver Section
LED Indicators
BURN_CTRL_1
BURN_CTRL_2
1k5
R16
R2
R1
D7
PWR
D4
LED
D3
EM Driver Section
Trigger Switches
Burn1
Burn2
BURN_CTRL_1
BURN_CTRL_2
TSA063G43–250
EM Driver Section
Power + Trigger Signals
J5
Molex Picoblade
or
Pico–Lock Connector
FM Board
Power + Trigger Signals
J6
10uF
C7
EM Driver Section
Microcontroller Conn.
J1
MCU_BURN_1
MCU_BURN_2
R24
R25
330
Current–limiting resistors prevent MCU damage
if buttons pressed.
FM Board – Burn Circuit
Decoupling Capacitors
C8
47nF
C9
100nF
C10
10uF
U2GND
Active High
3.3V or 5V control
3k
R29
R30
U2
BTS7030–2EPA
INO
DEN
IS
DSEL
IN1
NC
VS
OUT0
OUT1
R12
R13
Burn Resistors
Recommended
Resistor values:
4R7 @ 5V
36Ω @ 12V
R27
22k
R28
22k
R23
10k
R26
10k
R33
Pulldowns
R32
47
C11
100nF
C12
100nF
Reverse & ESD Protection
CubeSource
Sheet: /
File: Burnwing.kicad_sch
Title: Burnwing
Size: USLetter
Date: 2026–08–03
Rev: 3
KiCad E.D.A. 10.0.5
Id: 1/1